INCH-POUND
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20 August 1985

DETAIL SPECIFICATION

INTERCOMMUNICATION CONTROL UNIT

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and acceptance of an intercommunication control unit for use on military vehicles requiring selective communication between crew members.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-22750	- Coating, Epoxy, High Solids.
MIL-PRF-62488	- Amplifier, Audio Frequency.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

DEPARTMENT OF DEFENSE

MIL-STD-462	- Measurement of Electromagnetic Interference Characteristics, Test Methods for.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

12310226	- Control Unit Assembly.

(Copies of this drawing are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Materials. Unless otherwise specified, the materials selection shall be the prerogative of the contractor as long as all articles submitted to the Government fully meet the operating, interface, support and ownership, and operating environment requirements specified.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds all specified requirements, and promotes economically advantageous life cycle costs.

3.2.2 Corrosion resistance. All ferrous metal parts shall be of corrosion resistant type or treated for corrosion resistance.

3.3 Design and construction. The design, construction, and assembly of the control unit shall conform to Drawing 12310226. The control unit shall be painted Forest Green (color No. 34079) in accordance with FED-STD-595 or equivalent with an epoxy-polyamide chemical agent resistant coating per MIL-PRF-22750, type I, or equivalent.

3.4 Interface requirements. The assembly shall conform to the envelope dimensions, electrical interfaces, and mounting dimensions and locations as specified on Drawing 12310226.

3.4.1 Size and weight. The control unit shall not exceed 3.25 pounds (lb) (1.48 kilograms (kg)) in weight. The size shall not exceed the following dimensions: 4.5 inches (in.) long by 4.88 in. wide by 3.38 in. deep (11.43 centimeters (cm) by 12.39 cm by 8.58 cm).

3.4.2 Monitor switches. The control unit shall contain four independent monitor switches which will permit the crew member to monitor any combination of four radios.

3.4.3 Transmitter selector switch. The control unit shall contain a five position switch to permit selection of RADIO 1, RADIO 2, RADIO 3, RADIO 4, and Intercom only. As permitted by the radio transmission selector switch as specified in MIL-PRF-62488, this switch routes microphone audio and keying signals to the selected radio. Selection of a radio shall also permit monitoring of the receiver audio from the selected radio regardless of the monitor switch position.

3.4.4 Intercom cable assembly. The control unit shall interconnect with other audio system components by means of CX-4723/VRC cable. The 18 socket connector shown in figure 1 shall interface with the male connector on cable CX-4723/VRC.

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3.4.5 Audio interface. The control unit shall interface with a variety of audio accessories. Primary interface details are shown in figure 2. When handsets are used with the control units, they are connected to the control unit through J802 to allow radio transmission or to J803 to allow intercom transmission. When headsets are used with the control unit, they are connected to the control unit through J802 and J803 to allow both radio and intercom transmission. Monitoring of the intercom and radio shall not be affected by connection to either J802 or J803. The control unit shall interface with the distribution unit through either J801 or J804 to enable radio and intercom functions.

3.5 Operating requirements. Each control unit shall provide the following functional, operational, and performance capabilities.

3.5.1 Microphone amplifier.

3.5.1.1 Input impedance. The input impedance from pin D to pin A of J802 and J803 shall be 150 ±23 ohms (Ω) at 1000 Hertz (Hz).

3.5.1.2 Microphone to radio gain. The voltage gain from pin D of J802 and J803 to pins K, N, U and V of J801 and J804 shall be 110 ±10 volts (V) per V at 1000 Hz.

3.5.1.3 Radio output impedance. The output impedance at pins K, N, U and V of J801 and J804 shall be 1330 ±133 Ω at 1000 Hz.

3.5.1.4 Microphone to radio frequency response. The microphone to radio frequency response shall be flat ±2 decibels (dB) from 500 Hz to 3000 Hz.

3.5.1.5 Microphone to radio distortion. With a 7 millivolt (mV) root mean square (rms) 1000 Hz signal applied to the microphone input of J802 or J803, and output taken at pins K, N, U and V of J801 and J804, the total harmonic distortion shall be less than 2 percent (%).

3.5.1.6 Microphone to radio crosstalk. The voltage at any unselected output shall be at least 50 dB below the output of the selected radio output.

3.5.2 Radio to headset.

3.5.2.1 Radio to headset gain. The voltage gain from pins J, M, T and S of J801 and J804 to pin B or E at J802 shall be 140 ±30 V per V with a 500 Ω load at pin B or E and a frequency of 1000 Hz.

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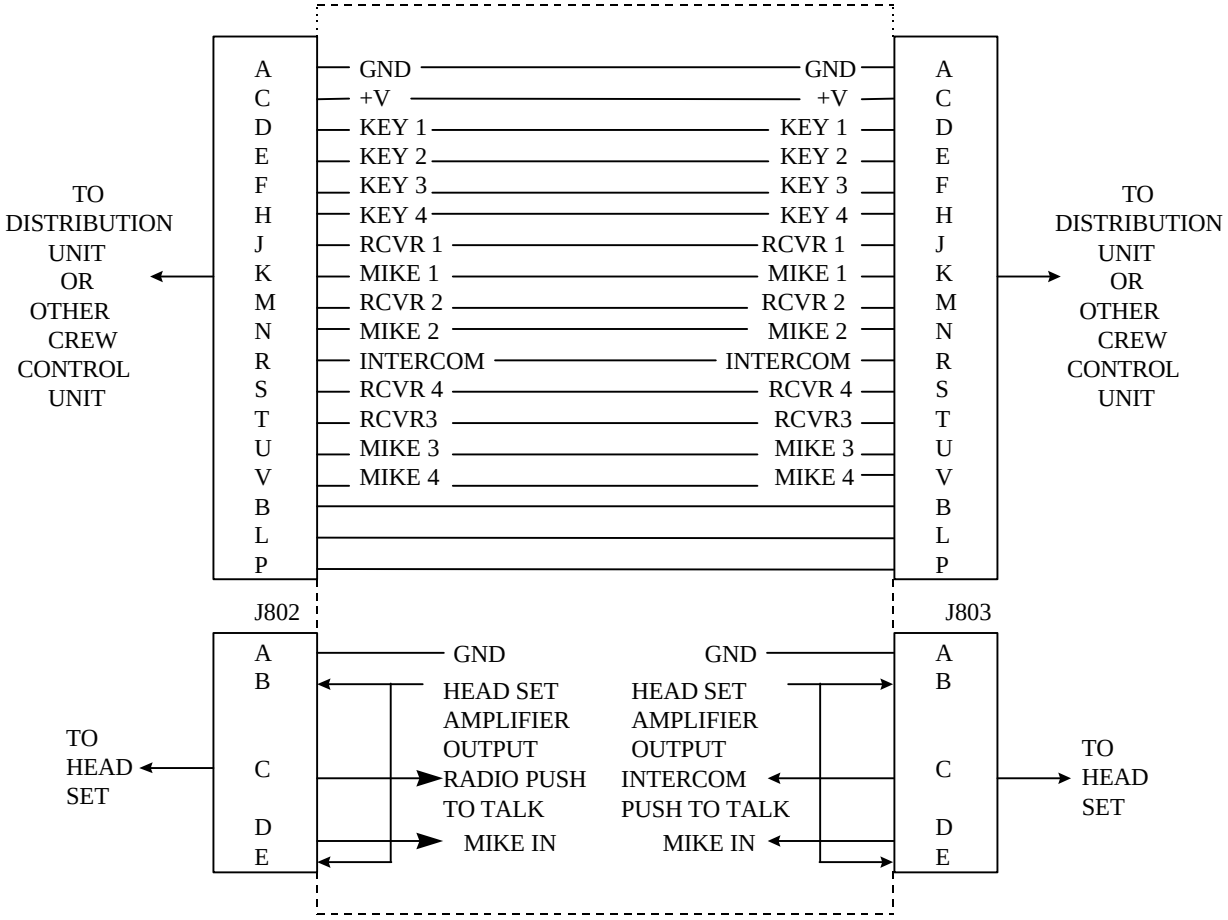


FIGURE 1. Control unit electrical interface.

3.5.2.2 Headset output power. The headset output shall be capable of delivering 200 milliwatts (mW) minimum into 500 Ω at 1000 Hz. The output shall not be damaged by operating into a short circuit.

3.5.2.3 Radio to headset frequency response. The radio to headset frequency response shall be as shown below with output terminated into 500 Ω.

Frequency	500 Hz	1000 Hz	3000 Hz
Output Level	±2 dB	0 dB (Reference)	±7 dB

3.5.2.4 Radio to headset distortion. The radio to headset total harmonic distortion shall be less than 5% with output delivering 200 mW rms into 500 Ω at 1000 Hz.

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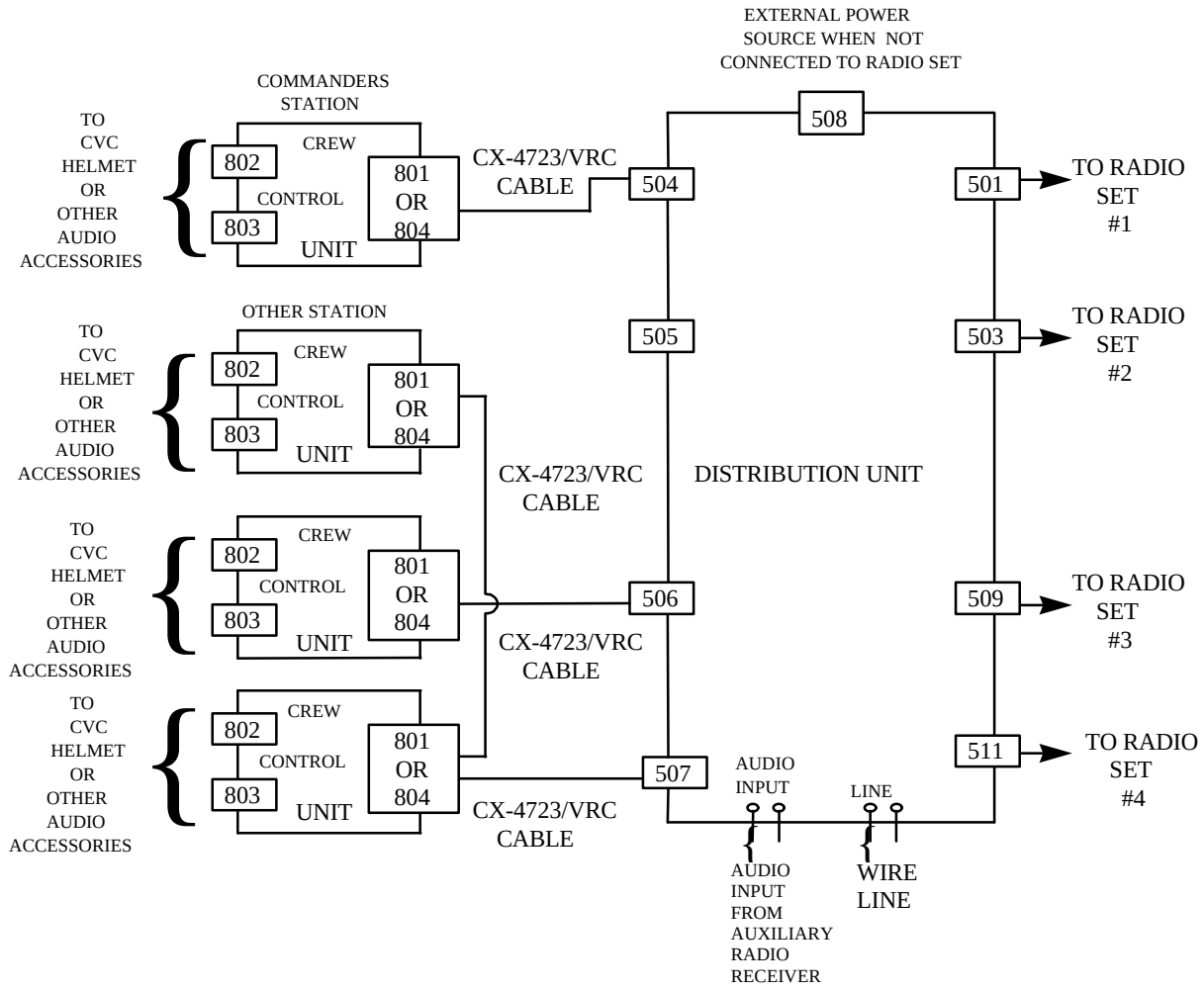


FIGURE 2. Control unit interface connections.

3.5.2.5 Radio to headset crosstalk. The voltage output at the headset from any unselected input shall be at least 50 dB below the output from a selected input.

3.5.3 Microphone to headset.

3.5.3.1 Microphone to headset gain. The voltage gain from the microphone input to headset output shall be 1610 ±430 V per V with the headset output terminated into 500 Ω and input frequency of 1000 Hz.

3.5.3.2 Microphone to headset frequency response. The microphone to headset frequency response shall be flat ±6 dB from 500 Hz to 3000 Hz.

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3.5.3.3 Microphone to headset distortion. The microphone to headset total harmonic distortion shall be less than 5% at 1000 Hz with the headset output delivering 200 mW rms into 500 Ω.

3.5.4 Microphone to intercom.

3.5.4.1 Microphone to intercom gain. The voltage gain from the microphone input to pin R of J801 and J804 shall be 4.29 ±0.43 V per V at 1000 Hz.

3.5.5 Volume control. This control shall permit the crew member to adjust the audio power provided to his headset. The volume control shall exhibit an audio taper characteristic and shall provide a power output variation of at least 45 dB.

3.5.6 Keying. With pin C open circuited and RADIO 1 selected, the resistance from pin D to pin A on J801 shall exceed 10 000 Ω. With pins A and C of J802 shorted, the resistance between pins D and A of J801 (RADIO 1 selected) shall be less than 2 Ω. It shall not be possible to key more than one radio simultaneously from any given control unit. The keying circuit line shall be capable of withstanding an open circuit voltage of 26 V direct current (dc).

3.6 Support and ownership requirements. Each control unit shall possess the following life cycle ownership characteristics.

3.6.1 Identification marking. Identification marking shall be permanent and legible and as a minimum, shall include:

a. Item name
b. Part number
c. Serial number
d. National Stock Number (NSN)
e. Manufacturer’s name and Identification (CAGE)
f. 24 Vdc
g. Date of manufacturer.

3.7 Operating environment requirements. Each control unit shall operate under the following environmental conditions without damage or loss of performance.

3.7.1 Temperature. The control unit shall operate as specified in 3.5.1 through 3.5.6 in ambient air temperatures of -25 to +160 degrees Fahrenheit (°F) (-32 to +71 degrees Celsius (°C)). The control unit shall withstand storage temperatures of -40 to +160°F (-40 to +71°C) for a duration of 6 hours.

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3.7.2 Low pressure (altitude). The control unit shall operate as specified in 3.5.1 through 3.5.6 at altitudes of 0 to 10 000 feet (ft) (0 to 3050 meters (m)). Non-operating, the control unit shall withstand exposure to altitude from 0 to 40 000 ft (0 to 12 200 m).

3.7.3 Humidity. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 during and after exposure to cyclic humidity conditions of 5% to 95% relative humidity at ambient temperatures of 90 to 145°F.

3.7.4 Fungus. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 after 90 days of exposure to conditions favorable to fungus growth.

3.7.5 Salt fog. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 following cyclic exposure to a 5% salt fog atmosphere for a minimum of four (2 wet and 2 dry) 24 hour periods.

3.7.6 Sand and dust. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 following exposure to blown sand particles of 0.0000039 to 0.039 in. (0.0001 to 1 mm) in diameter with predominant diameters between 0.0059 and 0.0118 in. (0.15 to 0.3 mm), and with velocity of 30 knots (55.6 kilometer per hour (km/hr)) to 35 knots (64.8 km/hr).

3.7.7 Water resistant. The control unit shall be spray tight, such that following a cleaning by hosing with water, it shall be capable of operating as specified in 3.5.1 through 3.5.6.

3.7.8 Explosive atmosphere. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 and not cause ignition of an ambient-explosive-gaseous mixture of air when operating in a fuel vapor-laden.

3.7.9 Vibration. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 while being subjected to vibration levels of 1.5 and 4.2 gravity (g) in frequency range of 5 to 500 Hz a sweep time of 15 times in each of three major axes, and shall evidence no permanently deformed, cracked, or broken parts.

3.7.10 Shock. The control unit shall be capable of operating as specified in 3.5.1 through 3.5.6 following the shock of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate encountered during normal operation.

3.7.11 Electromagnetic interference. When specified (see 6.2), electromagnetic interference (EMI) shall be in accordance with table I, and figures 3, 4, and 5.

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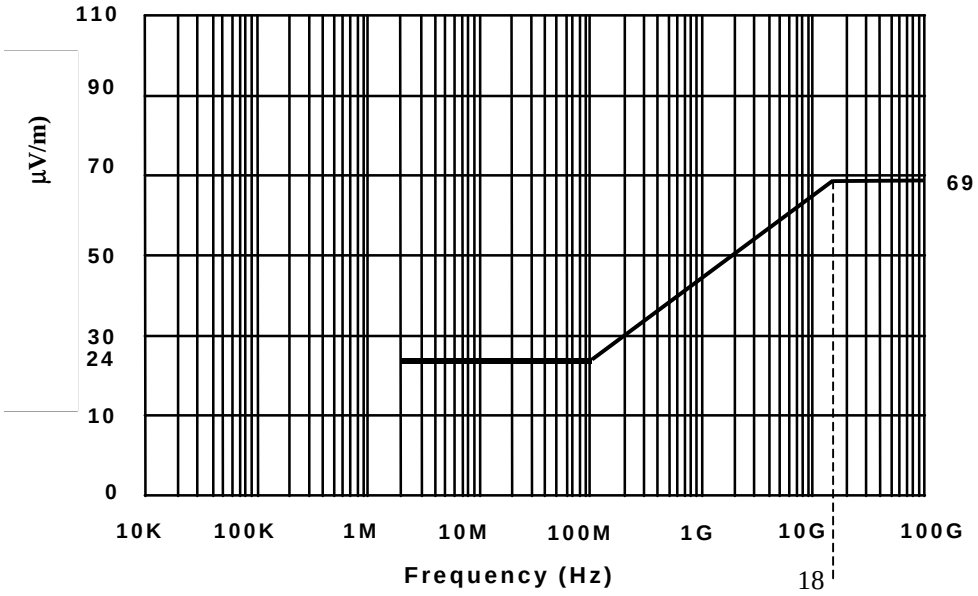


FIGURE 3. Radiated emission.

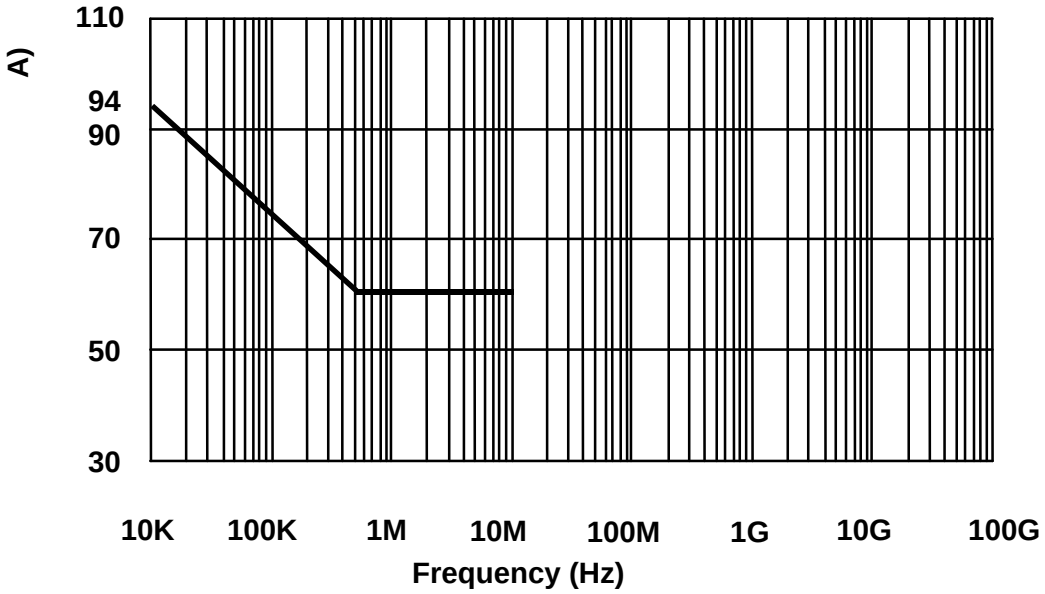


FIGURE 4. Conducted emission.

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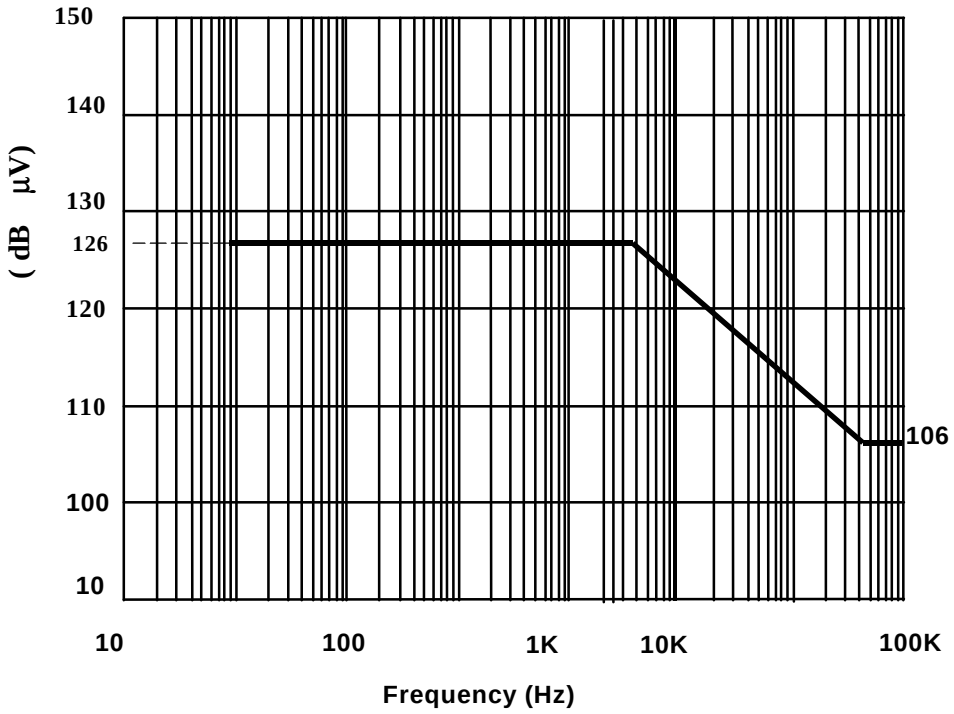


FIGURE 5. Conducted susceptibility.

TABLE I. Radiated susceptibility limits.

Frequency range	limit level (volts/meter)
10 kHz 1/ to 2 MHz 2/	20
2 MHz to 30 MHz	50
30 MHz to 1 GHz 3/	50
1 GHz to 18 GHz	50
18 GHz to 40 GHz	50

1/ kHz = kilohertz 2/ MHz = megahertz 3/ GHz = gigahertz

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
   1. Preproduction inspection (see 4.1.1.1).
   2. Initial production inspection (see 4.1.1.2).
b. Conformance inspections.

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4.1.1 First article inspection. First article inspection shall be performed on preproduction or initial production when required by the contract. When a first article inspection is required (see 6.2 and 6.3), it includes all verifications listed in table II.

4.1.1.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two intercommunication control units. Preproduction inspection shall consists of inspection as specified in table II.

4.1.1.2 Initial production inspection. Unless otherwise specified the Government shall select two control units, from the first ten production control units for initial production inspection. Initial production units shall be inspected as specified in table II.

4.1.2 Conformance inspections. Each control unit shall be subject to the examinations and tests specified in table II.

4.2 Verification methods. Acceptable verification methods included in this section are visual inspection, and measurement, sample tests, full-scale demonstration tests, simulation, modeling engineering evaluation, component properties analysis, and similarity to previously-approved or previous-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.2.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73.4 $\pm$18°F (23 $\pm$10°C)
b. Atmospheric pressure: 28.5 (+2, -3) in. of mercury (Hg)
   96.7 (+6.7, -10) kilopascals (kPa)).
c. Relative humidity: 50 $\pm$30 percent.

4.2.3 Materials, design and construction. The contractor’s records shall be inspected to provide proof that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards test reports and rating data. The control unit shall be visually inspected for evidence of dirt or grease build-up; chipped, scratched or peeling paint: rust: corrosion or incorrect finish.

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TABLE II. Verification methods.

Title	Requirement	Inspection	First article inspection	Conformance inspection
Materials	3.2 thru 3.2.2	4.2.3	X	
Design and construction	3.3	4.2.3	X	
Interface requirements:	3.4	4.2.4		
Size and weight	3.4.1	4.2.4	X	X
Monitor switches	3.4.2	4.2.4	X	X
Transmitter selector switch	3.4.3	4.2.4	X	X
Intercom cable assembly	3.4.4	4.2.4	X	X
Audio interface	3.4.5	4.2.4	X	X
Operating requirements:	3.5	4.2.5		
Microphone amplifier	3.5.1 thru 3.5.1.6	4.2.5.1 thru 4.2.5.1.6	X	X
Radio to headset	3.5.2 thru 3.5.2.5	4.2.5.2 thru 4.2.5.2.5	X	X
Microphone to headset	3.5.3 thru 3.5.3.3	4.2.5.3 thru 4.2.5.3.3	X	
Microphone to intercom	3.5.4 thru 3.5.4.1	4.2.5.4 thru 4.2.5.4.1	X	X
Volume control	3.5.5	4.2.5.5	X	X
Keying	3.5.6	4.2.5.6	X	
Support and ownership requirements:	3.6	4.2.6		
Identification and marking	3.6.1	4.2.6.1	X	X
Operating environment requirements:	3.7	4.2.7		
Temperature	3.7.1	4.2.7.1.1 & 4.2.7.1.2	X	
Altitude	3.7.2	4.2.7.2	X	
Humidity	3.7.3	4.2.7.3	X	
Fungus	3.7.4	4.2.7.4	X	
Salt fog	3.7.5	4.2.7.5	X	
Sand and dust	3.7.6	4.2.7.6	X	
Water spray	3.7.7	4.2.7.7	X	
Explosive atmosphere	3.7.8	4.2.7.8	X	
Vibration	3.7.9	4.2.7.9	X	
Shock	3.7.10	4.2.7.10	X	
Electromagnetic interference	3.7.11	4.2.7.11	X	

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4.2.4 Interface requirements verification. Verify the envelope dimensions and mounting interface by measurement and placement, and the electrical interface by connection and disconnection of cables and connectors, in the existing locations on various types of applicable vehicles. Perform an operating check, to insure that the control unit is fully compatible with all existing components.

4.2.5 Operating requirement verification. Except as otherwise specified herein, equipment performance shall be measured with a 56 Ω resistor in parallel with an 825 Ω and 0.27 microfarad series combination connected between pin R and signal ground in order to simulate system operation. Except as otherwise specified herein, gain measurements shall be made with equipment volume controls set to maximum.

4.2.5.1 Microphone amplifier.

4.2.5.1.1 Input impedance. The input impedance shall be measured from pin D to pin A of J802 and J803, and the input impedance shall be 150 ± 23 ohms at 1000 Hz.

4.1.5.1.2 Microphone to radio gain. With a 7 mV rms, 1000 Hz signal applied to pin D of J802 or J803 and output taken at pins K, N, U and V of J801 and J804 and the TALK switch in the appropriate position while pin C of J802 is grounded, each output voltage shall measure 0.77 V rms ±70 mV. This output corresponds to a gain of 110 ±10 V per V per at 1000 Hz.

4.2.5.1.3 Radio output impedance. The output impedance shall be measured at pins K, N, U and V of J801 and J804 with respect to pin A of J801 or J804. The headset output shall be capable of delivering 200 mW minimum into 500 Ω at 1000 Hz.

4.2.5.1.4 Microphone to radio frequency response. With a constant amplitude, variable frequency input signal applied to pin D of J802 or J803, the rms voltage at pins K, N, U and V of J801 and J804 with the proper output selected shall be as shown below.

Frequency	500 Hz	1000 Hz	3000 Hz
Output Level	±2 dB	0 dB (Reference)	±2 dB

4.2.5.1.5 Microphone to radio distortion. With a 7 mV rms, 1000 Hz signal applied to pin D of J802 or J803, the distortion measured at pins K, N, U and V of J801 and J804 shall be less than 2% at each output.

4.2.5.1.6 Microphone to radio crosstalk. With a 7 mV rms, 1000 Hz signal applied to pin D of J802 or J803 and output taken at pins K, N, U and V of J801 and J804 and the TALK switch in the appropriate position while pin C of J802 is grounded, the voltage output at the headset from any unselected input shall be at least 50 dB below the output from a selected input.

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4.2.5.2 Radio to headset.

4.2.5.2.1 Radio to headset gain. With a 70 mV rms, 1000 Hz signal applied to pins J, M, T and S of J801 and J804 and output taken at pin B or E of J802 with a 500 Ωload and the appropriate channel selected, the output voltage shall measure 9.8 V rms ±2.1 V. This output corresponds to a gain of 140 ±30 V per V.

4.2.5.2.2 Headset output power. With sufficient input voltage applied to pin J of J801 or J804 and pin B or E of J802 terminated into a 500 Ω load, the output power shall be at least 200 mW rms at pin B or E of J802. The amplifier shall not be damaged by operating into a short circuited load at maximum output.

4.2.5.2.3 Radio headset frequency response. With a constant amplitude, variable frequency input signal applied to pins J, M, T and S of J801 or J804 and output taken at pin B or E of J802 and J803 with a 500 Ω load and the appropriate channel selected, the proper output shall be as shown below.

Frequency	500 Hz	1000 Hz	3000 Hz
Output Level	±2 dB	0 dB (Reference)	±7 dB

4.2.5.2.4 Radio to headset distortion. With a 70 mV rms, 1000 Hz signal applied to pins J, M, T and S of J801 or J804, the distortion shall be measured at pin B or E of J802 and J803. The radio to headset total harmonic distortion shall be less than 5% with output delivering 200 mW rms into 500 Ω at 1000 Hz.

4.2.5.2.5 Radio to headset crosstalk. With a 70 mV rms, 1000 Hz signal applied to pins J, M, T and S of J801 or J804 and output taken at pin B or E of J802 with a 500 Ω load and the appropriate channel selected, the voltage at the output from any unselected input shall be at least 50 dB below the output from a selected input.

4.2.5.3 Microphone to headset.

4.2.5.3.1 Microphone to headset frequency response. With a 7 mV rms, 1000 Hz signal applied to pin D of J802 or J803 and pin C of J803 grounded with output at pin B or E of J802 with a 500 Ω load, the output voltage shall measure 11.27 V rms ±3 V. This output corresponds to a gain of 1610 ±430 V per V.

4.2.5.3.2 Microphone to headset frequency response. With a constant amplitude, variable frequency input signal applied to pin D of J802 or J803 and output taken at pin B or E of J802 and J803 with a 500 ohm load and pin C of J803 grounded, the output shall be as shown below.

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Frequency	500 Hz	1000 Hz	3000 Hz
Output Level	±6 dB	0 dB (Reference)	±6 dB

4.2.5.3.3 Microphone to headset distortion. With a 7 mV rms, 1000 Hz signal applied to pin D of J802 or J803, the distortion measured at pin B or E of J802 and J803 shall be less than 5%.

4.2.5.4 Microphone to intercom.

4.2.5.4.1 Microphone to intercom gain. With a 7 mV rms, 1000 Hz signal applied to pin D of J802 or J803 and output taken at pin R of J801 and J804, the output voltage shall measure 30 ±3 mV. This output corresponds to a gain of 4.29 ±0.43 V per V.

4.2.5.5 Volume control. The test of 4.2.5.2.2 shall be performed, except the volume control shall be adjusted to minimum. Output power shall be at least 45 dB below the level observed in the test of 4.2.5.2.2.

4.2.5.6 Keying. With pin C of J802 open circuited, and RADIO 1 selected, the resistance from pin D to pin A of J801 shall be measured. With pins A and C of J802 shorted, the resistance between pin D and pin A of J801 (RADIO 1 selected) shall be measured The keying circuit shall be opened and the keying line subjected to a power supply voltage of 26 Vdc via an inductive-resistive device (relay coil or equivalent) resistance of 56 Ω ±10% and inductance of 10 millihenries ±20%. The keying circuit shall be exercised and no degradation shall occur.

4.2.6 Support and ownership requirements verification.

4.2.6.1 Identification and marking. The presence of the required markings on the control unit shall be inspected. After performing all environmental tests in 4.2.7, the control unit markings shall be re-inspected for readability.

4.2.7 Operating environment requirements verification.

4.2.7.1 Temperature test.

4.2.7.1.1 Low temperature. The control unit shall be tested in accordance with method 502.3, procedure I (storage) and procedure II (operation) of MIL-STD-810, or equivalent (see 4.2.1). The storage temperature shall be -40°F (-40°C) and storage duration shall be 6 hours. The operating temperature shall be -25°F (-32°C). Following the storage test, and during the operation test the unit shall be subjected to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

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4.2.7.1.2 High temperature. The control unit shall be tested in accordance with method 501.3, procedures I and II of MIL-STD-810, or equivalent (see 4.2.1). The control unit shall operate in ambient air temperature of +160°F (+71°C) and storage duration shall be 6 hours. Following the storage test, and during the operation test the unit shall be subjected to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

4.2.7.2 Low pressure (altitude). The control unit shall be tested in accordance with method 500.3, procedures I and II of MIL-STD-810, or equivalent (see 4.2.1), at pressures equivalent to an altitude of 10 000 ft (3050 m) for operating equipment and 40 000 ft (12 000 m) for non-operating equipment. Following the non-operating test, and during the operation test the unit shall be subjected to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

4.2.7.3 Humidity. The control unit shall be tested in accordance with method 507.3, procedure II of MIL-STD-810, or equivalent (see 4.2.1). The conditions shall conform to figure 507.3-2, cycle 5, except the humidity limits shall be 5 and 95% and the number of test cycles shall be ten. Equipment shall not be moved from the humidity chamber for measurements, and the power shall not be left on after measurements have been completed. If any unit fails to meet the tests of 4.2.5.1 through 4.2.5.6 during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the unit still fails to meet the tests of 4.2.5.1 through 4.2.5.6 during cycling, or fails subsequently, it does not pass the tests. In addition, if the unit fails to meet applicable requirements after final conditioning and adjustment, it does not pass the test. The test shall be performed with the cover off.

4.2.7.4 Fungus. The control unit shall be subjected to 90 days of the fungus test specified in method 508.4, procedure I of MIL-STD-810, or equivalent (see 4.2.1). The parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall be sprayed. The control unit shall be subjected to the test in this open condition. Following the fungus exposure, the control unit shall show no evidence of viable fungus or corrosion, and shall be subjected to and pass the tests of 4.2.5.1 through 4.2.5.6. In lieu of a test, when approved by the Government contract authority, certification, with supporting data, may be provided attesting that the item is constructed of materials which will not support fungus growth.

4.2.7.5 Salt fog. The control unit shall be tested in accordance with method 509.3, procedure I of MIL-STD-810, or equivalent (see 4.2.1) for a minimum of four (2 wet and 2 dry) 24 hour periods except the salt fog fallout shall be 25 lbs/acre/year. Following the test, the unit shall be subjective to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

4.2.7.6 Sand and dust. The control unit shall be tested in accordance with MIL-STD-810, method 510.3, procedure I, or equivalent (see 4.2.1), except the temperature during test step 5 shall be limited to 145°F (63°C) and the sand particles shall be

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0.00039 to 0.039 in. (0.01 to 1 mm) in diameter with predominant diameters between 0.0059 and 0.0118 in (0.15 to 0.3 mm), and with winds greater than 30 knots (55.6 km/hr). The dust shall have a density of 0.00000037 pounds per cubic foot (lb/ft^3) (0.000000006 grams per cubic centimeter (cm^3), with particle size 0.0000039 to 0.00039 in. (0.0001 to 0.01 mm) in diameter, and winds at 35 knots (64.8 km/hr). Following the test, the unit shall be subjective to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

4.2.7.7 Waterspray. The control unit shall be tested in accordance with the spray tight requirements as specified in table III, when mounted by its normal mounting means. Following the test, the unit shall be subjective to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

TABLE III. Water repellent tests using hose.

Enclosure	Rate of water flow (gal/min)	Form of water flow	Nozzle type	Water head (ft)	Approx. distance from nozzle to enclosure (ft)	Time of test (min)
Spray-tight	65	Solid stream	Solid stream nozzle, 1 inch diameter tip opening	As required	10	5

4.2.7.8 Explosive atmosphere. The control unit shall be tested in accordance with method 511.3, procedure I of MIL-STD-810 or, equivalent (see 4.2.1). Following the test, the unit shall be subjective to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

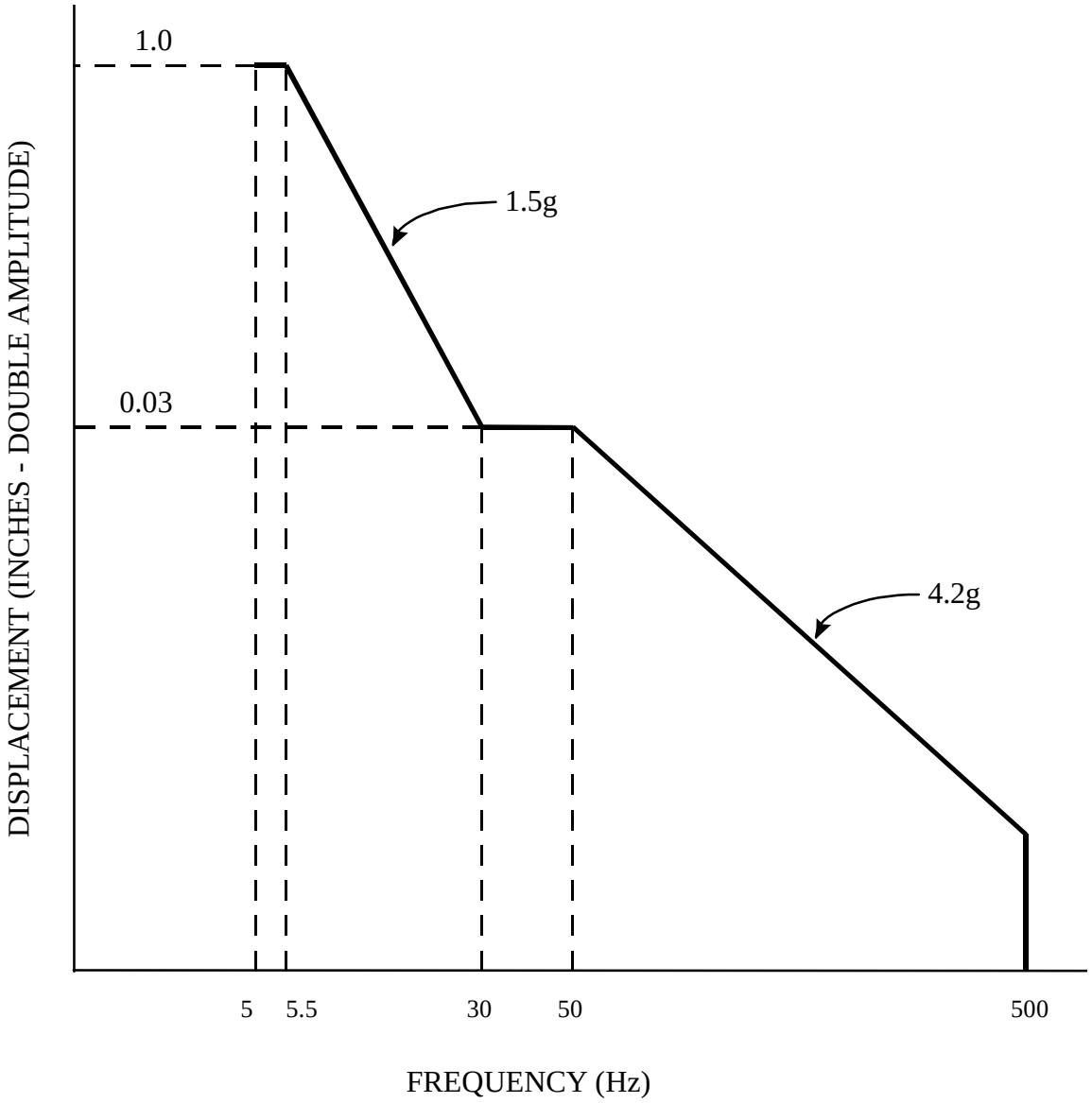
4.2.7.9 Vibration. The control unit shall be tested in accordance with MIL-STD-810, method 514.4, category 8, procedure I, or equivalent (see 4.2.1), with the test level as specified in figure 7. The control unit shall be operated during the test by injecting a 1000 Hz tone into all radio input channels. The output power shall be monitored and shall remain constant. The output tone shall be monitored and there shall be no repetitive breakup or garbling. Channels shall be periodically reselected. A sinusoidal cycling time of 30 minutes per 1000 miles or a maximum of 3 hours is required. Sweep time is 15 minutes from 5 to 500 to 5 Hz. Following the test, the unit shall be subjective to and shall pass the tests of 4.2.5.1 through 4.2.5.6.

4.2.7.10 Shock ballistic. The test shall be conducted on an appropriate shock testing machine. A control unit shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the unit. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the unit under test is permissible. Upon completion of the tests and

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before repairs are made, the unit shall be subjective to and shall pass the tests of 4.2.5.1 through 4.2.5.6 and any physical damage shall be minor only.

4.2.7.11 Electromagnetic interference. The electromagnetic interference shall be tested in accordance with the applicable provisions of MIL-STD-462, or equivalent (see 4.2.1).



NOTE: All curves shall be extended to 2 Hz when item resonances below 5 Hz are expected.

FIGURE 6. Vibration test curve.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This control unit is intended to be used in the FIST Vehicle or other military vehicles with communication intensive tasks. The control unit is military unique because it must be able to function and operate satisfactorily and reliably under all the environmental extremes that exceed commercial requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. When first article inspection is required (see 3.1, 4.1.1 and 6.3).
d. If EMI suppression is required (see 3.7.11).
e. If inspection conditions should be other than as specified (see 4.2.2).
f. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

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6.4 Subject term (key word) listing.

Amplifier
Communication
Headset
Microphone
Personnel communication
Radio
Receivers
Selective personnel communication
Signal

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 5820-0201)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-62481A(AT)	2. DOCUMENT DATE *(YYMMDD)* 980930

3. DOCUMENT TITLE

INTERCOMMUNICATION CONTROL UNIT

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340
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